

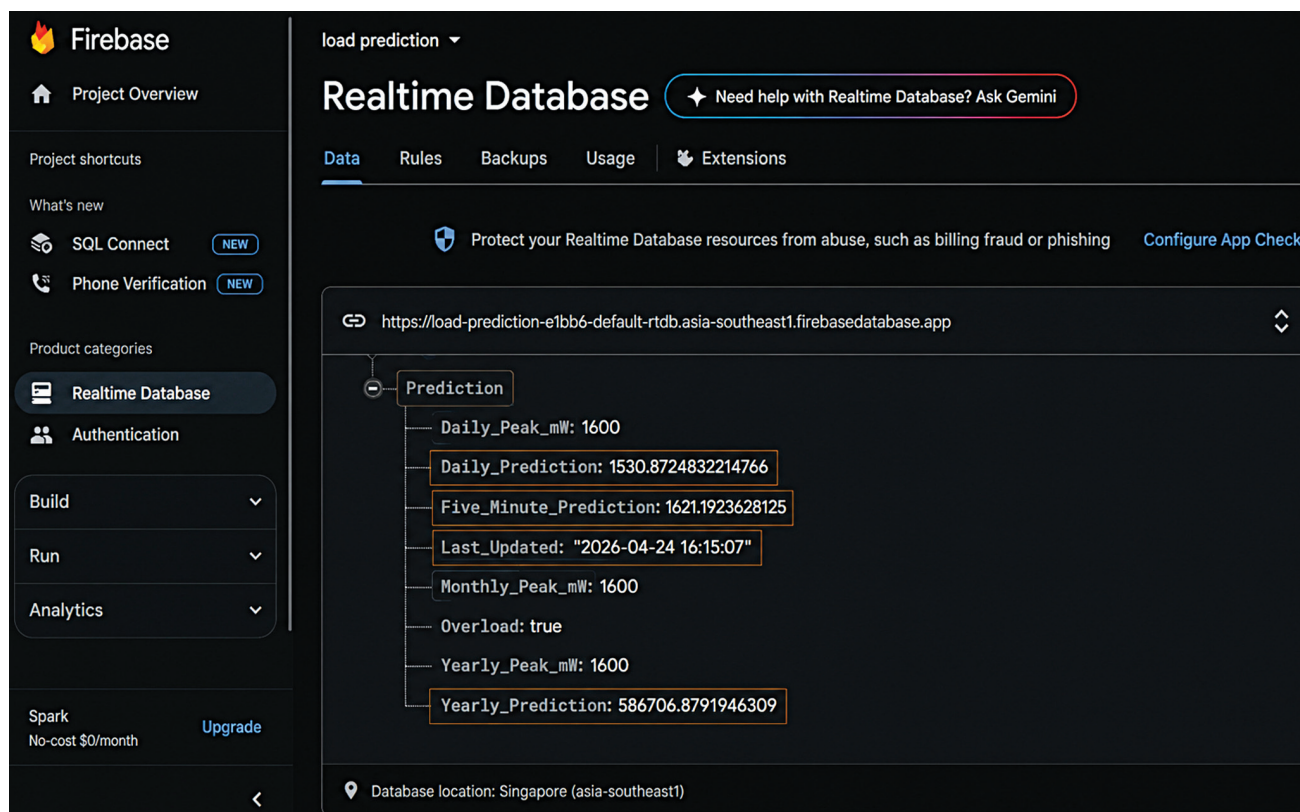


Fig. 9: Firebase real-time database

Real-time Database

2. Click Create Database
3. Select the database location
4. Choose Start in test mode (recommended during development) and click Enable

5. Copy the real-time database URL, as it will be required in the Arduino sketch

Fourth step: Enable authentication (optional)

If your project uses user authentication (recommended for production), enable anonymous authentication.

1. Go to Build→Authentication and click Get Started
2. Open the Sign-in Method tab
3. Enable anonymous sign-in and save the changes

EFY note. For simple ESP32 testing, only the real-time database, database URL, and web API key are required. Authentication can be added later to improve security.

Python code

Python is used for AI analytics and

usage pattern prediction. TensorFlow and Scikit-learn are used for AI model training and deployment, while Microsoft Excel is used to visualise data in spreadsheet form.

In case Python is not already installed in the computer, download and install the latest version from <https://www.python.org/downloads/>. Next, install the required Python libraries using pip, including NumPy, Scikit-learn, TensorFlow, Pandas, Matplotlib, and the Firebase Admin SDK. After installation, import these modules into the Python program. Configure the paths for the Firebase authentication key (JSON) file, the Excel file used to store the retrieved data, and the output directory. Also, specify the Firebase real-time database URL created during the Firebase configuration.

The Python program then retrieves sensor data from Firebase, processes and visualises the data, and applies AI algorithms, such as LSTM, using TensorFlow to predict

future load patterns. The measured and predicted values are then stored in an Excel spreadsheet for further analysis. Fig. 5 shows the Python code snippet.

Construction and testing

First, upload the source code to the ESP32 and assemble the circuit on a breadboard. Refer to the prototype in Fig. 2 and the circuit diagram in Fig. 4 before making the connections.

After completing the hardware setup, run the Python program on the computer. Ensure that all required Python libraries are installed and that the service account key, API key, and Firebase real-time database URL are correctly configured. Place the service account key file in the specified directory, power up the ESP32, and execute the Python program. If the setup is correct, the console output appears as shown in Fig. 6.

Fig. 6 shows the successful execution of the AI-based substation